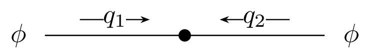
Correlator Diagram Examples

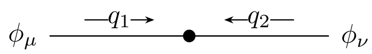
This file is the broad package tour. It starts with the shared controls, then moves through the available topologies. The focused regression files cover the edge cases in more detail.

1. Basic Entry Points

Default 2-point correlator:

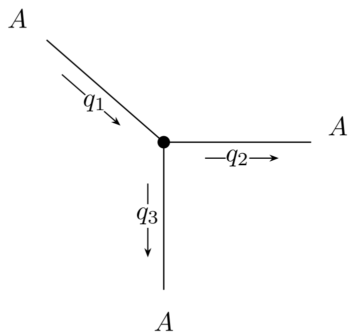


2-point correlator with generated momentum labels and field labels:

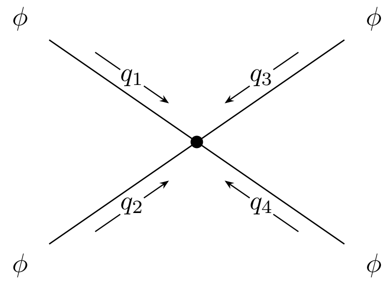


Default 3-point and 4-point correlators:

3-point vertex

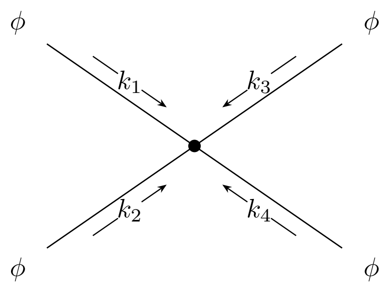


4-point contact

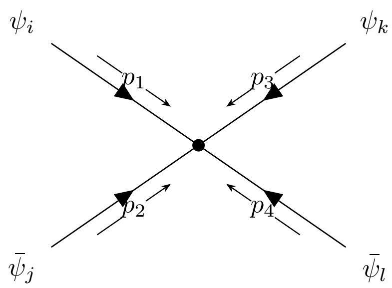


2. External Labels And Line Styles

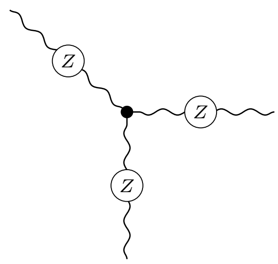
Automatic k_i momentum labels:



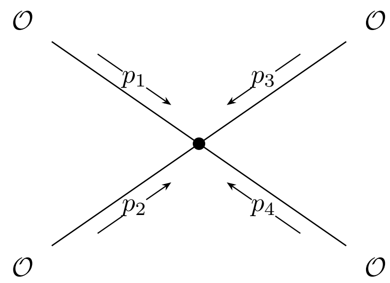
Manual external labels and a global fermion line style:



Circled marker on an EW-boson propagator:

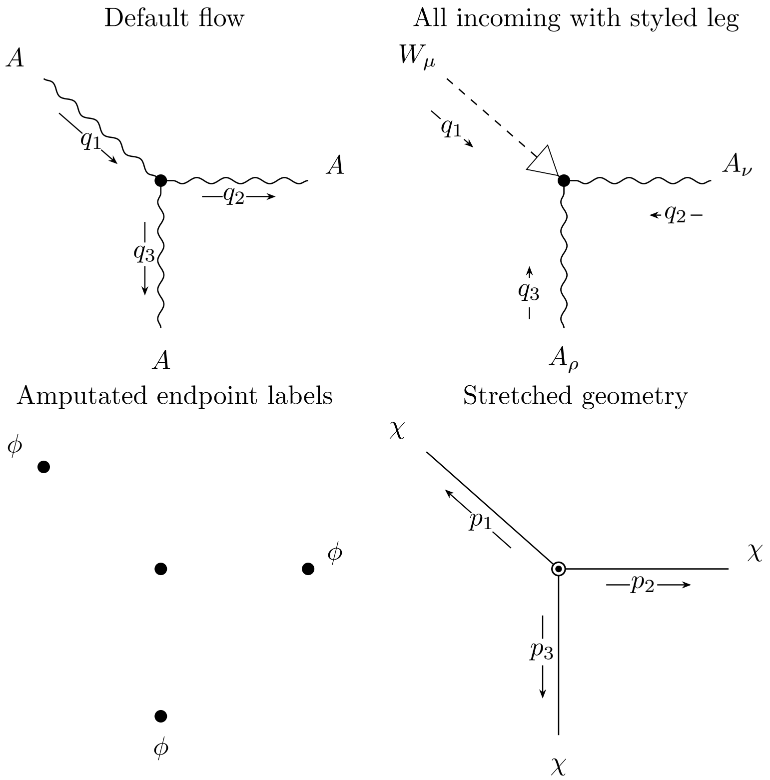


Four-point contact blob hidden:

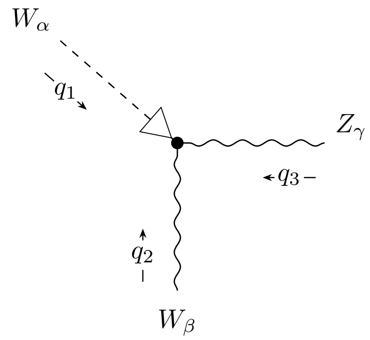


3. Three-Point Vertices

The 3-point slot order is upper-left, right, bottom. The default momentum flow is upper-left incoming and the other two outgoing.

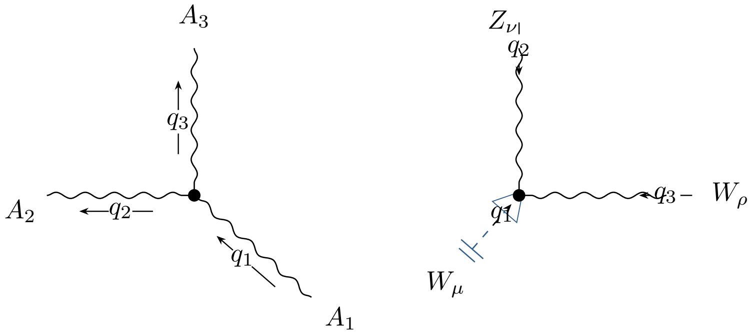


Opt-in outward momentum placement gives crowded triple vertices more room:



Three-point orientation presets rotate the slot geometry without changing slot order. The special slot-1 external leg can also be shortened and styled as `epspol`, which adds the double endcap.

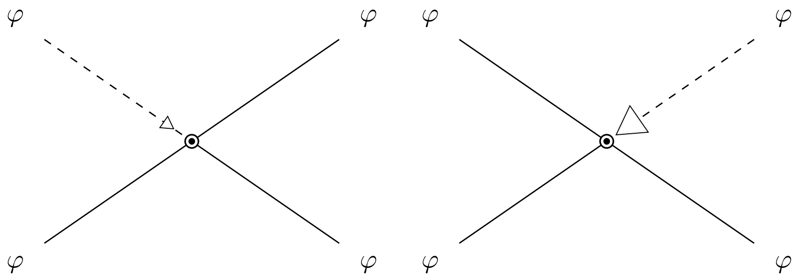
Left orientation | Up orientation with short `epspol`



4. Scalar-Polarized Legs

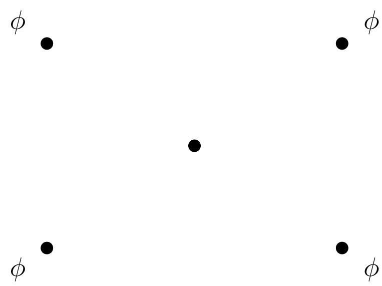
Scalar-polarized legs are dashed scalar lines with an open marker. The marker size follows the propagator-arrow sizing controls.

Default scalar arrow | Larger scalar arrow

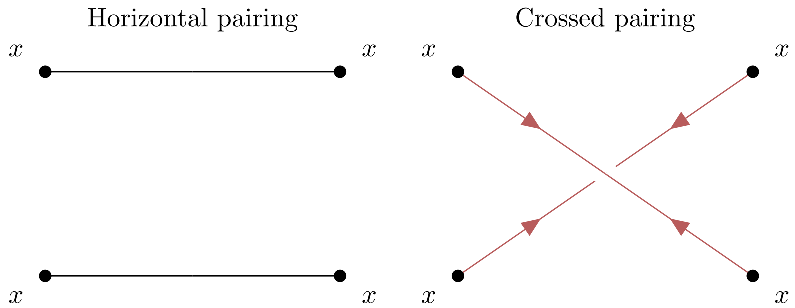


5. Contact Pairings And Amputation

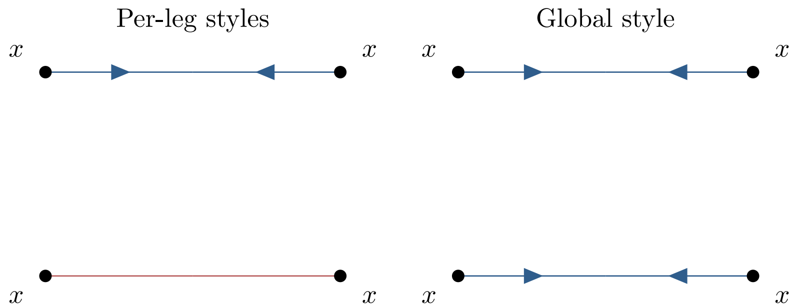
Amputated external legs with endpoint vertices:



Pairing controls for contact-style 4-point objects:

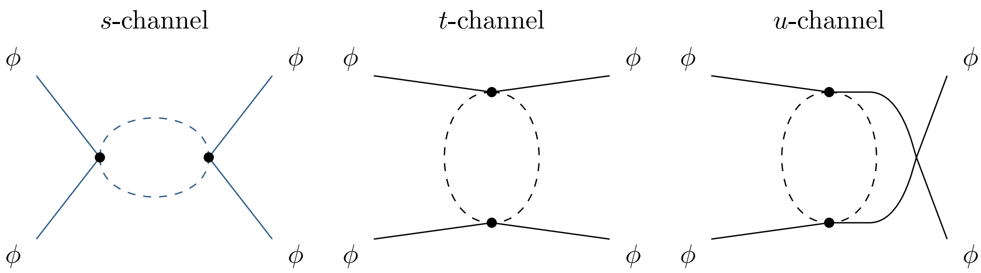


Per-leg color/style overrides versus a global line color:

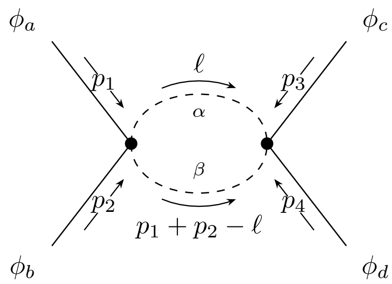


6. One-Loop $s/t/u$ Channels

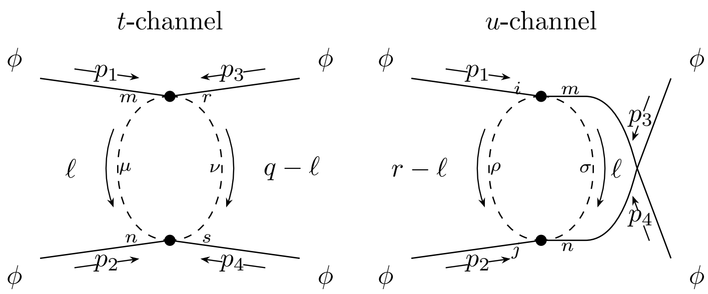
The channel wrappers are fixed-topology versions of `\FourPointCorr`.



Loop-channel momentum and contracted internal indices:



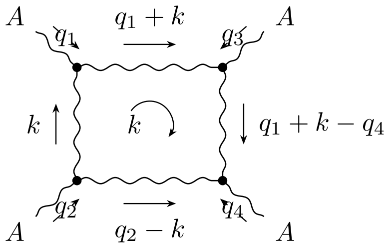
Separate vertex and propagator indices on the two internal lines:



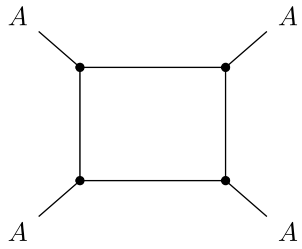
7. One-Loop Box Topology

The box topology is a four-point loop with clockwise edge momenta. For this topology the external slots follow the canonical box order: slot 1 is upper left, slot 2 is lower left, slot 3 is upper right, and slot 4 is lower right.

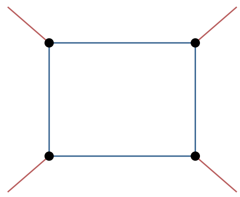
Default wavy box



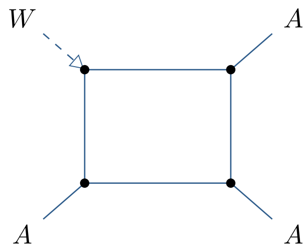
No momentum labels



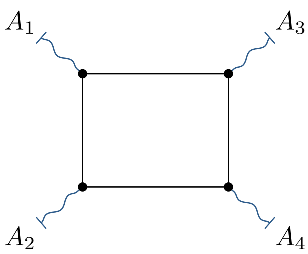
Colored box



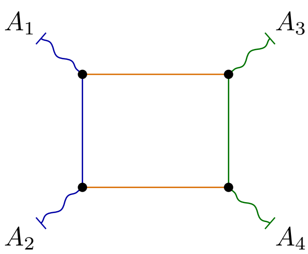
Mixed exterior styles still use the usual slot order:



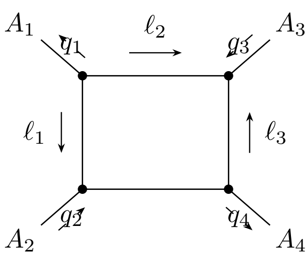
Physical-polarization external legs add EW-boson endcaps while keeping the internal box styling independent:



Grouped box colors can split the diagram into left, middle, and right regions:



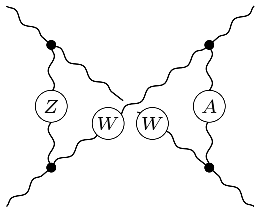
External and internal box momentum arrows can be flipped independently:



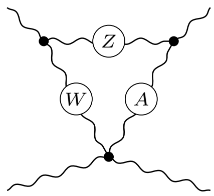
8. Cross-Box And Triangle-Contact Edge Styles

Cross-box diagrams accept the box edge line keys as a compact way to style the two side edges and the two crossed diagonals.

Cross-box circ edge styles



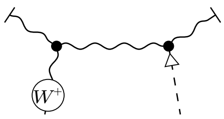
Triangle-contact internal styles



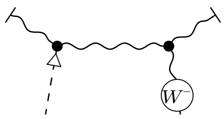
9. Vertex-Identity Half-Box Helpers

The half-box helpers are focused vertex-identity topologies. The `halfpropleft` and `halfpropright` styles angle the top stubs, add top endcaps, and pair one lower circled EW-boson half-propagator with one scalar-polarized leg.

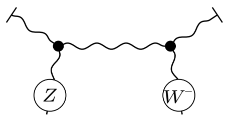
`halfpropleft`



`halfpropright`



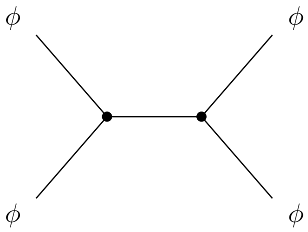
circ markers only



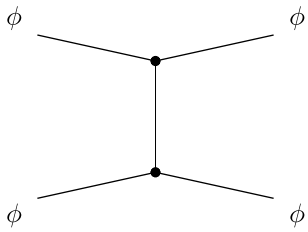
10. Tree-Level Exchange Channels

Tree-level exchange channels use the same external slots as the loop channels, but their internal propagator uses the older `channel-*` key family.

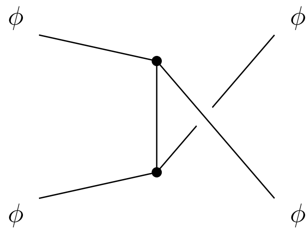
s-tree



t-tree

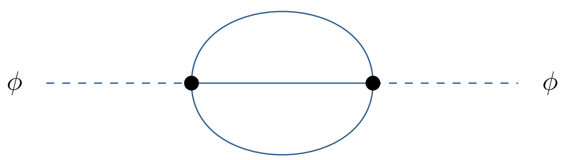


u-tree



11. Sunset Self-Energies

Minimal sunset diagram:



Sunset diagram with explicit loop labels and contracted internal indices: